From Real-World Problems to Research Questions, Hypothesis and Objectives

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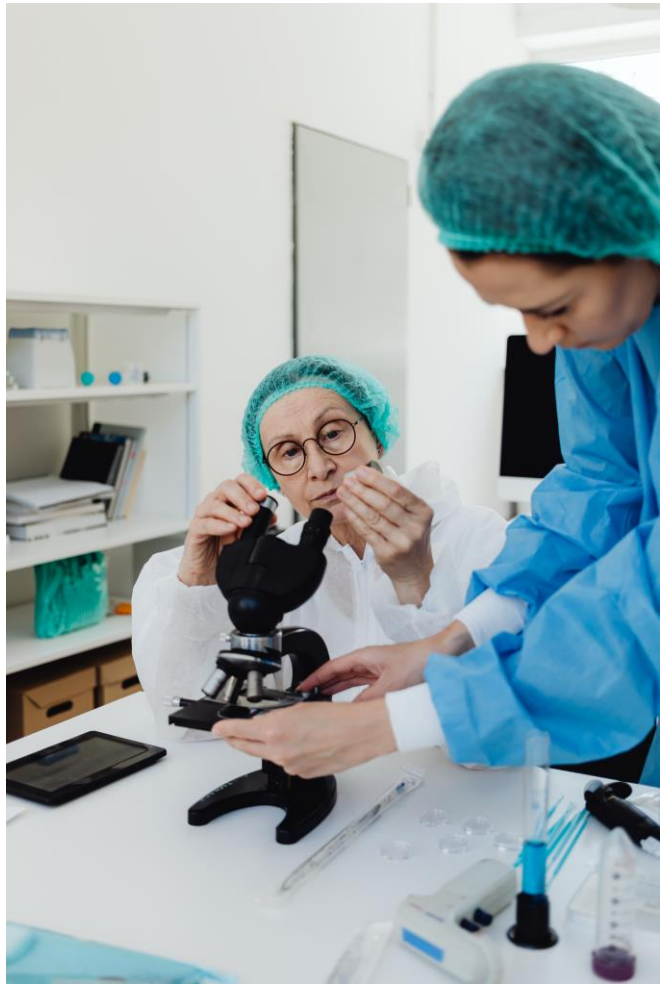


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Research has to be:

- Feasable: Addressed, tackled
- Interesting: Meaningful to research, to community
- Novel: Brings new insights
- Ethical: Must respect ethical principles
- Relevant: Research should matter



Order from chaos



Jackson Pollock: Number 5

The world is complex and messy. There are different ways to see this reality. There is also different ways in which we han pose a question. A scientist's objective when presenting a question is to be as precise as possible and to be able to objectively mesure the answer.

Specificity and **measurability**.

PICO framework

P	Problem	What is the research object?
I	Intervention	What treatment/experiment are you proposing?
C	Comparison	How do alternative hypothesis or methods compare?
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Problem: Real-world problem

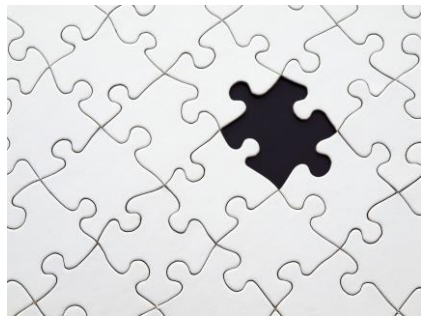
“Students are stressed.” is a real world problem.
Not yet a scientific question.

Academic workload may contribute to student stress, but the relationship is not fully understood.

Real world problem -> Question





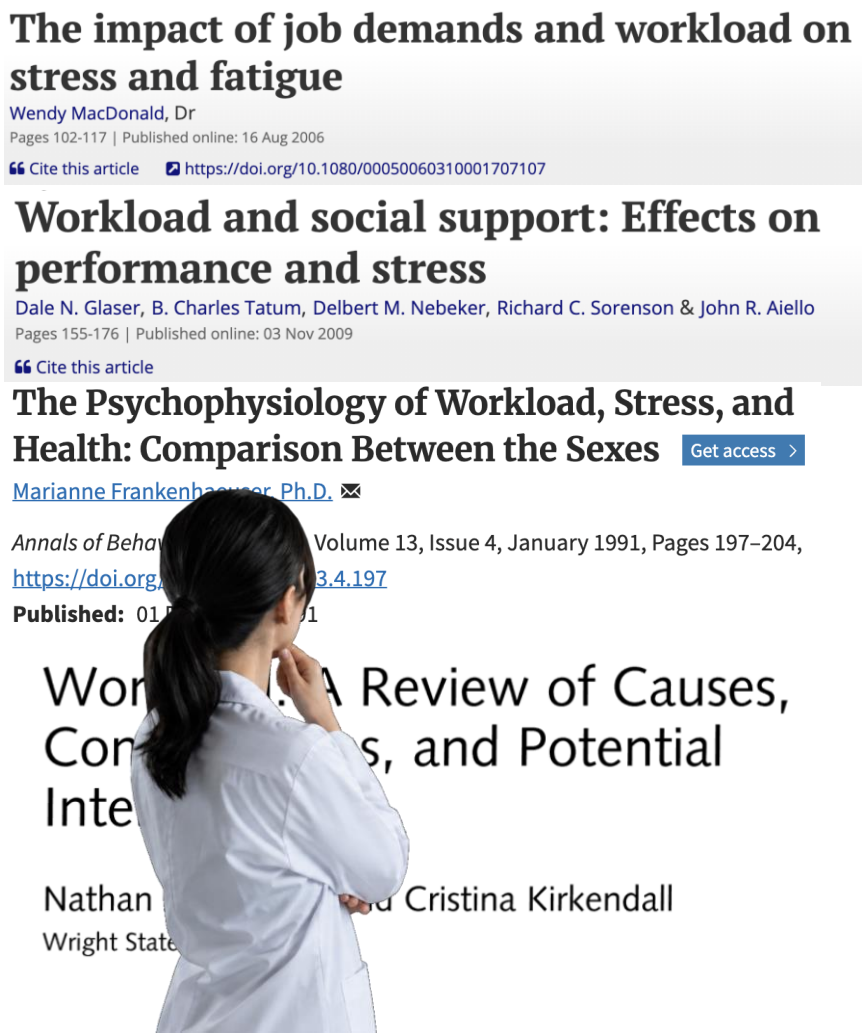


Has workload been associated with stress?
In which populations?
How strong is the evidence?
Are findings consistent?
What remains unclear?

Research gap



Problem: What do we already know?



Problem: Undergraduate students are stressed.
Gap: We don't sufficiently understand how minfulness-based stress reduction apps help.

"**In undergraduate college students experiencing high stress (P),** does the use of a mindfulness-based stress reduction app over a 6-week period (I) compared to no intervention (C) lead to a reduction in self-reported stress levels (O)?"

Intervention: From question to objective

Research object:
Undergraduate college students experiencing high stress.



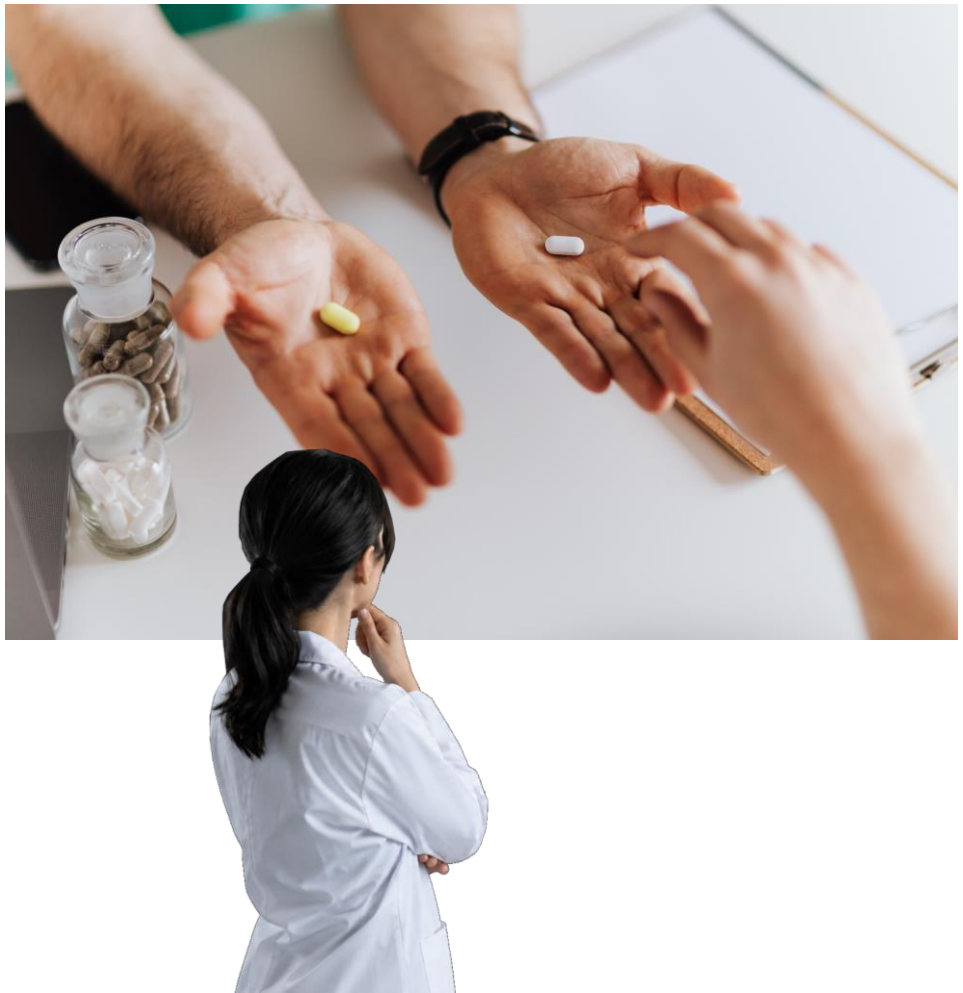
Objective
To estimate the association between academic workload and perceived stress among first-year students.

General objective
To assess the association between academic workload and stress.

Specific objectives
1. Measure academic workload.
2. Measure perceived stress.
3. Estimate the association between the two.
4. Assess whether the association differs across relevant groups.

Intervention: The methods

Intervention = the exposure, treatment, diagnostic strategy, or action being evaluated



💊 **Treatment:** Drug A vs standard therapy
🏃 **Behavioural:** Exercise programme
🩺 **Procedure:** Laparoscopic vs open surgery
🧠 **Psychological:** Cognitive behavioural therapy
🧪 **Diagnostic:** MRI vs CT
🏥 **System-level:** Nurse-led vs physician-led follow-up
🌍 **Public health:** Vaccination campaign / screening programme

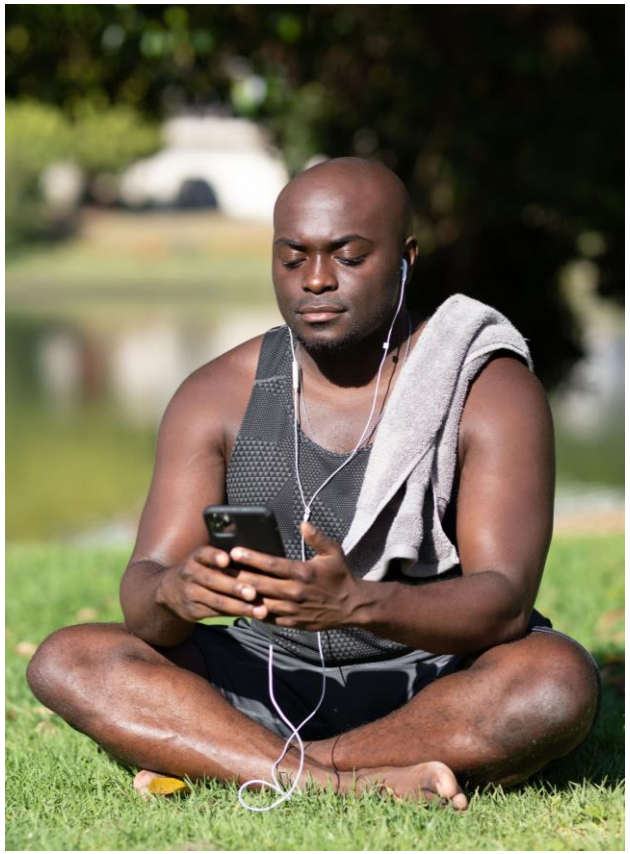
The intervention should be described precisely enough that others can know **what was actually done**.

- *What intervention/exposure am I interested in?*
- *Who receives it?*
- *What does it involve?*
- *For how long / how often?*
- *In what setting?*

Intervention: The methods

Avoid vague interventions	→	**Define the intervention operationally**
Exercise	→	150 min/week of moderate-intensity aerobic exercise
Antibiotics	→	Amoxicillin 1 g orally 3× daily for 7 days
Surgery	→	Laparoscopic cholecystectomy at week 4 of tumor detection

"In undergraduate college students experiencing high stress (P), does **the use of a mindfulness-based stress reduction app over a 6-week period (I)** compared to no intervention (C) lead to a reduction in self-reported stress levels (O)?"



Intervention = WHAT + HOW + DOSE/INTENSITY + DURATION + SETTING

Comparison: Alternative and null hypothesis



A hypothesis is a testable prediction based on existing knowledge or theory.

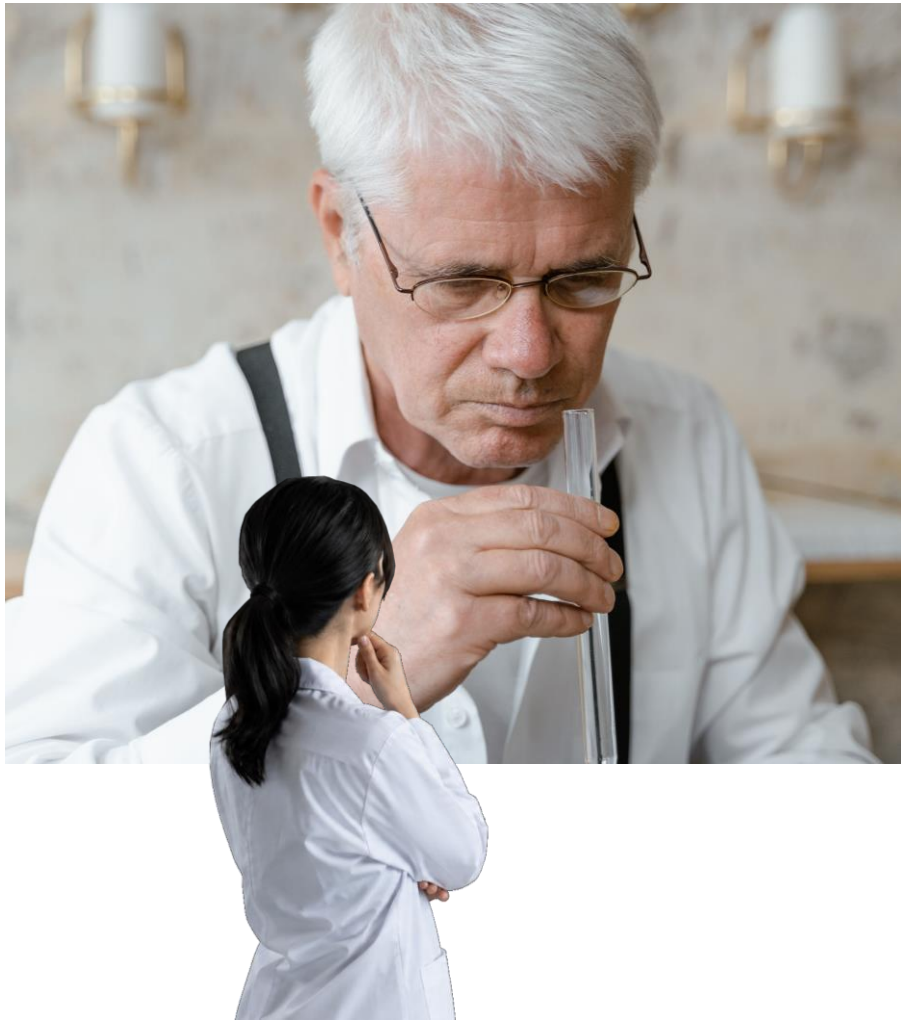
Research question:
Is workload associated with stress?

Hypothesis:
Students reporting higher academic workload will report higher levels of stress.

Comparing different models.

Comparing with no treatment.

Comparison: Blinding



Placebo effect.

Open-Label (Unblinded): All participants, clinicians, and researchers know the treatment being given.

Single-Blind: Only the participants are kept unaware of their treatment assignment (usually receiving a real treatment or a placebo).

Double-Blind: Both the participants and the treating clinicians or data collectors do not know who receives the experimental treatment or the control.

Triple-Blind: The participants, clinicians/data collectors, and the outcome adjudicators or data analysts are all kept unaware of the treatment groups until the study ends

Comparison: Not every research question need a hypothesis



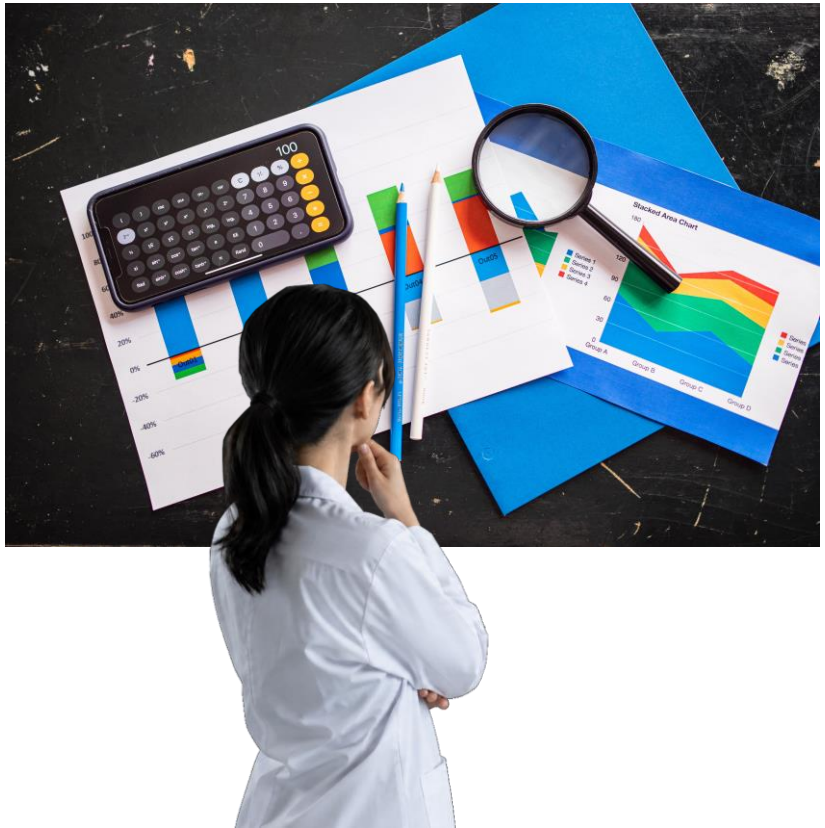
Descriptive
What is the prevalence of stress among students?
→ No hypothesis necessarily required.

Associational
Is workload associated with stress?
→ Hypothesis makes sense.

Exploratory
How do students describe their experience of academic stress?
→ An exploratory question may be more appropriate than a formal hypothesis.

Outcome: What do we expect to see?

Outcome = the measurable effect that matters to the patient, clinician, or health system



♡ **Clinical:** mortality, stroke, myocardial infarction

Disease-related: blood pressure, HbA1c, tumour recurrence

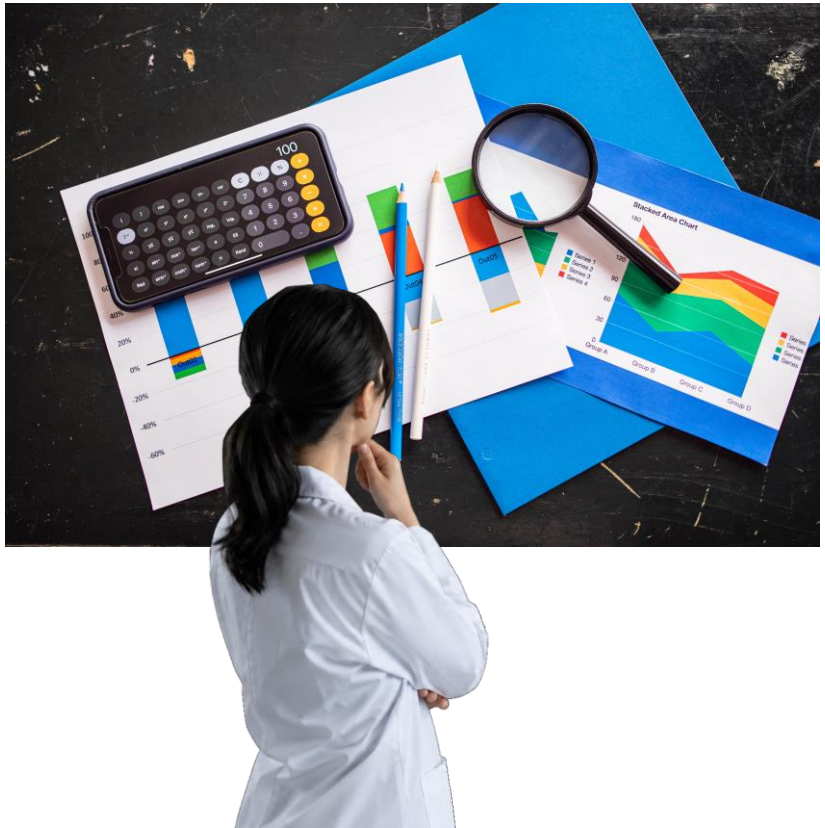
Patient-reported: pain, quality of life, functional status

⚠ **Safety:** adverse events, complications, hospitalisation

Healthcare: length of stay, readmission, healthcare costs

Outcome: What do we expect to see?

Avoid: “Does the intervention work?”
Ask: “What specific outcome will tell us whether it works?”



A good outcome should be:

- **Relevant:** matters to patients/clinical decision-making
- **Specific:** clearly defined and measurable
- **Time-bound:** *when* will it be assessed?
- **Objective or validated:** measured using a reliable method

"In undergraduate college students experiencing high stress (P), does the use of a mindfulness-based stress reduction app (I) compared to no intervention (C) **lead to a reduction in self-reported stress levels over a 6-week period (O)?"**

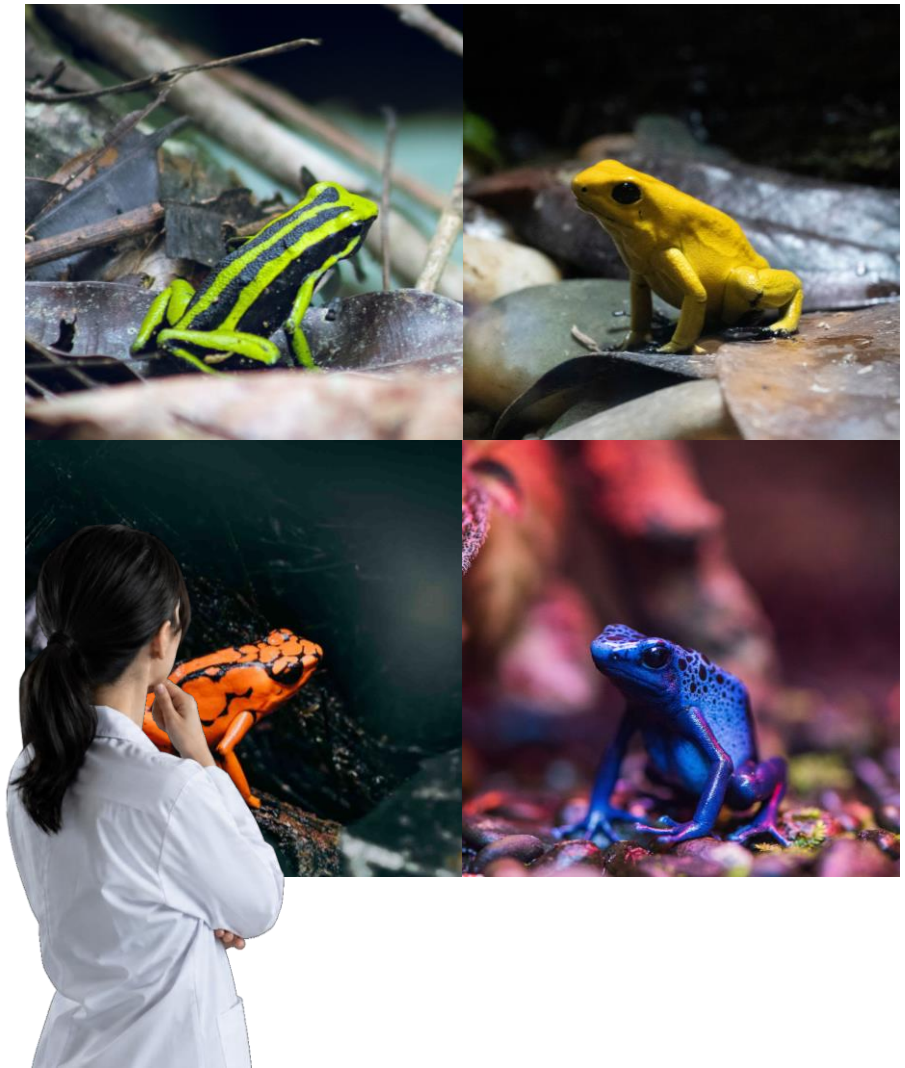
The scientific question

"In **undergraduate college students experiencing high stress (P)**, does the use of a **mindfulness-based stress reduction app (I)** compared to **no intervention (C)** lead to a **reduction in self-reported stress levels over a 6-week period (O)**?"



- **P (Patient / Population):** Undergraduate college students experiencing high stress or anxiety.
- **I (Intervention):** Mindfulness-based stress reduction app.
- **C (Comparison):** No intervention (standard college life) or traditional campus counseling services.
- **O (Outcome):** Reduced self-reported stress levels, improved sleep quality, or better academic performance.

Another case



Real world problem:
“Poison dart frogs develop poison only in the wild”

P (Patient / Population / Problem): Poison dart frogs (*Dendrobatidae*) raised in captive environments.

I (Intervention / Exposure): A wild-simulated diet consisting of alkaloid-rich prey for 2 years (e.g., specific wild beattles).

C (Comparison): A standard captive diet (e.g., vitamin-dusted fruit flies and pinhead crickets).

O (Outcome): The synthesis and accumulation of defensive skin toxins (alkaloids).

In **captive-raised poison dart frogs (P)**, does feeding a **wild-simulated, alkaloid-rich diet over a period of 2 years (I)** compared to a **standard fruit fly diet (C)** result in the **development of toxic skin secretions (O)**?

Your turn to use the PICO framework:

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